SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CASE STUDY

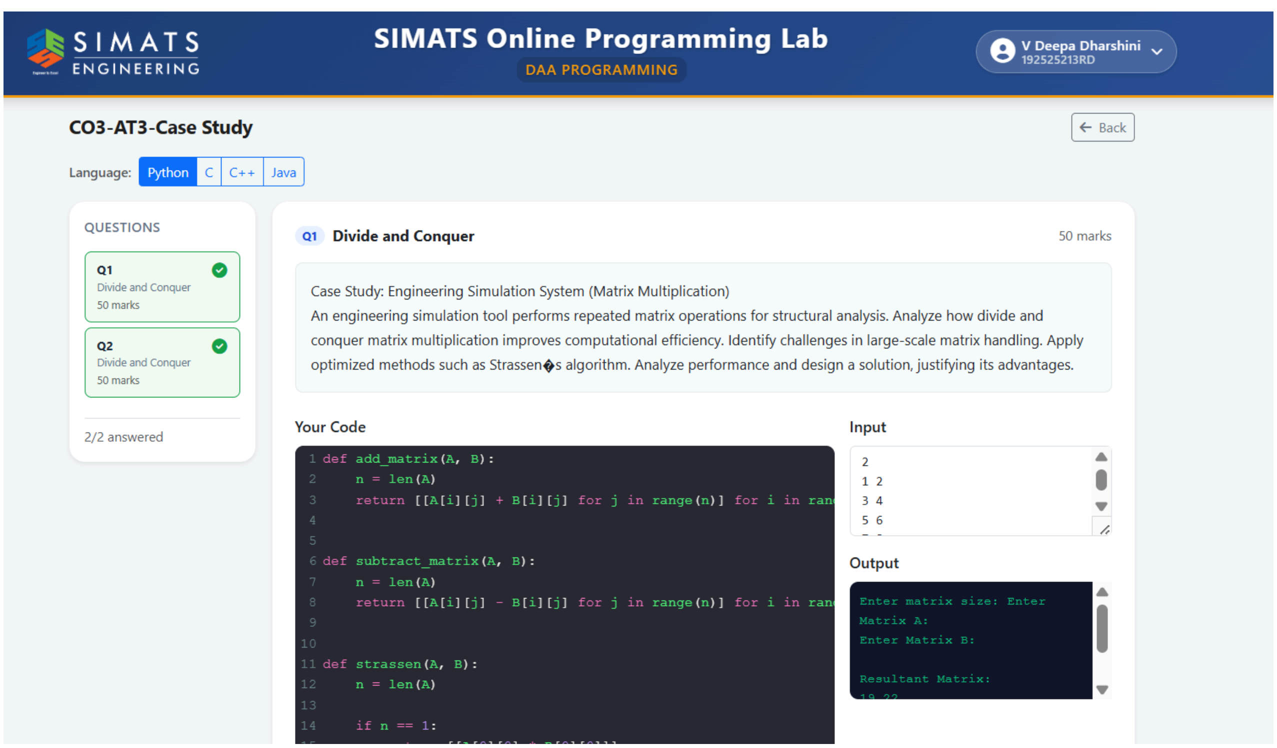
Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	**CO3:** Articulation (BL4, BL5)	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	V Deepa Dharshini	**Reg. No.:**	192525213
Section / Batch:	B.Tech AIML(2nd year)	**Date:**	18-08-2026

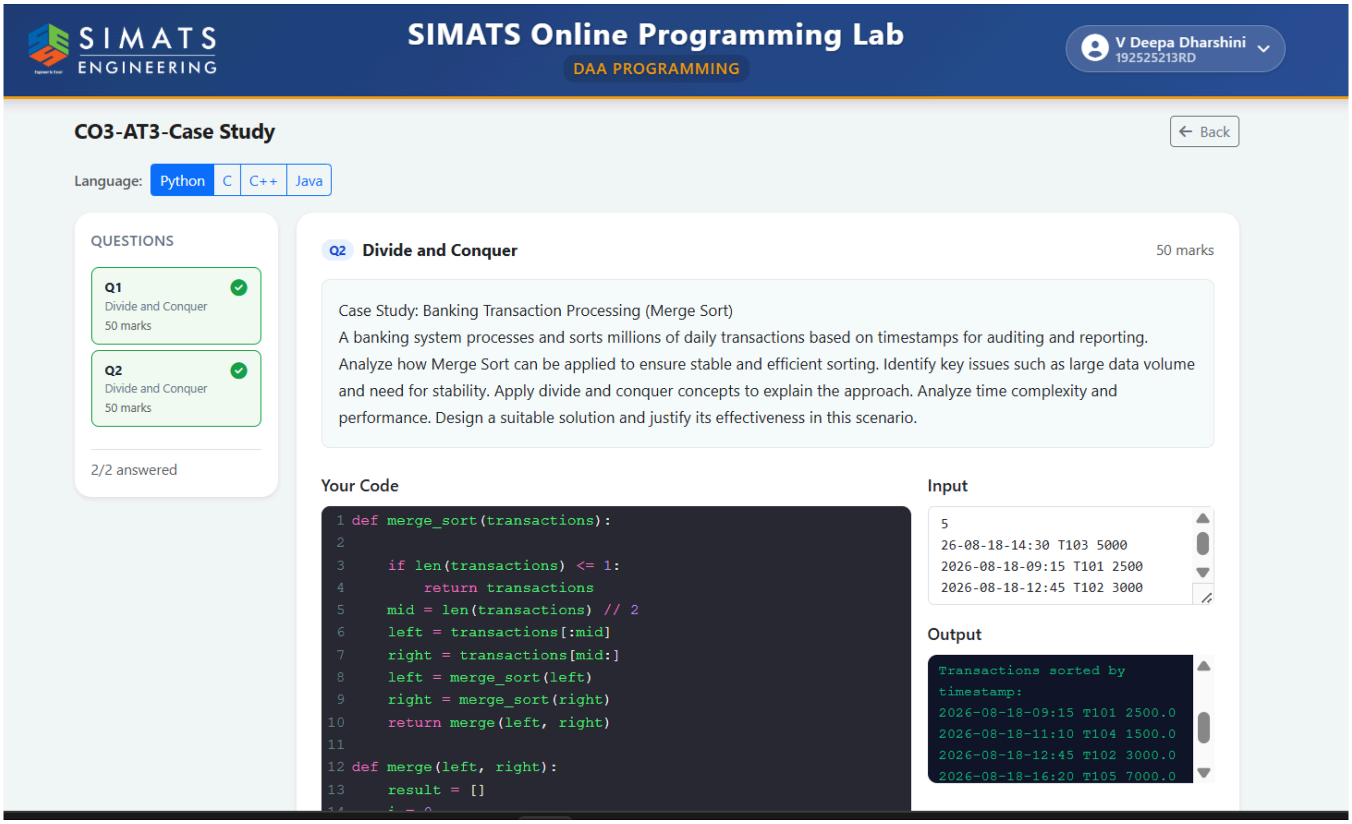
Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identifies all key issues, in-depth analysis, Innovative, feasible solutions with strong justification

Question Type	Description	Questions	Total Marks
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1 – Q2	100

SECTION A – CASE STUDY





Code of Conduct

I V Deepa Dharshini (192525213) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Deepa

RUBRICS FOR CALCULATION

Case Study					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Mark Evaluation:

Case Study	Understanding of case (25)	Applications of concepts (25)	Analysis (20)	Solution Design (20)	Clarity (10)	Total Marks (Each - 50)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Name: ____________________ Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________